

DEPARTMENT OF Electrical Engineering
UNIVERSITY OF ENGINEERING & TECHNOLOGY, PESHAWAR
JALOZAI CAMPUS

Course: Computer Programming (2nd Semester)

Assignment: Assignment No. 1 (Spring-2026)

Student Name: MUHAMMAD YAHYA AFRIDI

Registration No :BF25NWELE0755

i. Variable Declaration :

```
#include <iostream> using
namespace std;
int main()
{
    // Defining these as double since electrical values often have decimals
    double voltage, current, resistance;    return 0;
```

ii. Display

Message :

```
#include <iostream> using
namespace std;
int main()
{
    // Just a simple print statement to welcome the user
    cout << "Welcome to Object Oriented Programming in C++!";
    return 0; } iii. Read
```

Age :

```
#include <iostream> using
namespace std;
int main() {    int age;    cout << "Please enter your age: ";
cin >> age; // Saving the user input into the age variable
return 0;
```

iv. Temperature

Check :

```
#include <iostream> using
namespace std;    int
main() {    int
temperature = 105; //
Testing with
```

```

a value over 100
if
(temperature
> 100) {
cout <<
"Overheating
detected!";
// This only
runs if it's
too hot
}      return
0;
}

```

v. Area of a Circle :

```

#include <iostream> using
namespace std;  int
main() {      double r =
5.0, area;
    // Using the constant PI value given in the assignment
area = 3.14159 * r * r;      cout << "The area of the circle
is: " << area;      return 0;

```

} vi. Print x, y, and

z :

```

#include <iostream> using
namespace std;
int main() {      int x = 10, y =
20, z = 30;
    // Printing them all out with commas in between      cout
<< x << ", " << y << ", " << z;      return 0;

```

} vii. Sum of Voltage and

Current :

```

#include <iostream> using namespace std;  int
main() {      double voltage = 12.0, current = 0.5,
total;      total = voltage + current; // Adding them
together      return 0;

```

} viii. Division by Zero

Guard :

```

#include <iostream> using
namespace std;  int
main() {      int
denominator = 0;

```

```

    // Safety check to make sure we don't divide by zero
    if (denominator == 0) {          cout << "Error: Division
by zero";          }          return 0;
}

```

ix. Increment and

Print :

```

#include <iostream> using
namespace std;    int
main() {          int count =
10;

    // Incrementing first then printing the new value
    cout << ++count;          return 0;

}

```

x. Read Three Floats :

```

#include <iostream> using namespace std;    int
main() {          float a, b, c;          cout << "Enter
three floating-point numbers: ";          cin >> a >> b
>> c; // Getting all three at once          return 0; }

```

xi. Total Resistance

(Series) :

```

#include <iostream> using namespace std;    int
main() {          double r1, r2, r3, total;          cout <<
"Enter values for 3 series resistors: ";          cin
>> r1 >> r2 >> r3;          total = r1 + r2 + r3; //
Series is just the sum          cout << "Total Series
Resistance: " << total;          return 0; }

```

xii. Total Resistance

(Parallel) :

```

#include <iostream> using namespace std;    int
main() {          double r1, r2, r3, total;          cout
<< "Enter values for 3 parallel resistors: ";          cin
>> r1 >> r2 >> r3;

    // Using the reciprocal formula for parallel circuits          total
= 1.0 / (1.0/r1 + 1.0/r2 + 1.0/r3);          cout <<
"Total Parallel Resistance: " << total;          return 0;
}

```

xiii. Voltage Division Rule

(VDR) :

```

#include <iostream> using
namespace std; int main()
{
    // Picking some standard values for the circuit          double
R1 = 50, R2 = 100, V_source = 15, VR2;
}

```

```

        // VDR formula: (Source Voltage * R2) / Total Resistance
VR2 = (V_source * R2) / (R1 + R2);      cout << "The voltage across
R2 is: " << VR2;      return 0;
}

```

xiv. Current Division Rule (CDR) :

```

#include <iostream> using
namespace std;
int main() {
    // Selecting resistors and total current      double
R1 = 10, R2 = 40, I_total = 2.5, IR2;
    // CDR formula: (Total Current * R1) / (R1 + R2)
IR2 = (I_total * R1) / (R1 + R2);      cout << "The current
through R2 is: " << IR2;      return 0;
}

```

xv. Semester GPA

Calculation :

```

#include <iostream> using
namespace std;
int main() {      double g1, g2, g3, c1, c2, c3, gpa;      cout << "Enter
GPA and credits for 3 subjects: ";      cin >> g1 >> c1 >> g2 >> c2 >> g3 >>
c3;      // Multiplying each grade by its credits and dividing by total
credits      gpa = (g1*c1 + g2*c2 + g3*c3) / (c1 + c2 + c3);      cout <<
"My semester GPA is: " << gpa;      return 0;
}

```